

Silicoin: A Two-Asset, Isolated Multi-Layer Settlement Network

Jonathan Miller
silicoinddev@proton.me

Abstract

Single-token blockchain architectures struggle to reconcile conflicting economic goals: serving as a scarce store of value while also functioning as a liquid medium of exchange. Silicoin resolves this dichotomy through a native dual-token economy comprising Gold (a store of value with a tail emission) and Cash (a medium of exchange balanced by market dynamics). Rather than pegging to an external asset, Cash floats freely but maintains internal economic equilibrium through opposing supply pressures. A constant transaction burn fee (initially modelled at 5%) drives deflationary upward pressure, while issuing network bonds exerts expansionary downward pressure. This stabilisation method deliberately avoids vulnerable third-party oracles and gameable algorithmic loops. To scale, Silicoin employs Layer Development Kits (LDKs) to generate a hierarchical tree of independent child blockchains. By locking funds at the root network (L0), users can bridge assets to LDKs for localised, high-speed execution. This architecture isolates decentralised applications, custom tokens, and exchanges, preventing root-network congestion. Because child networks completely lack token-minting privileges, Sybil attacks on LDKs are economically neutralised. Finally, the entire Silicoin ecosystem is secured against future cryptographic threats using ML-DSA-65 post-quantum signatures from day one.

Contents

1	Introduction: The Structural Dilemma	3
2	Layer 0: The Dual-Token Economy	3
2.1	The Nakamoto Foundation	3
2.2	Silicoin Gold: Store of Value	4
2.3	Silicoin Cash: Medium of Exchange	4
3	Scaling via Layer Development Kits (LDKs)	5
3.1	State Compartmentalisation	6
3.2	Pristine Collateral and Local Execution	6
3.3	Cryptographic Checkpoints	6
3.4	LDK Lifecycle and Keep-Alives	7

3.5	Nested Hierarchy and Fee Autonomy	7
3.6	Scripting Expandability	8
4	Security: Post-Quantum Cryptography	8
5	Limitations & Future Work	8
6	Conclusion	9
	References	10

1 Introduction: The Structural Dilemma

Decentralised networks face four unresolved design conflicts:

1. **The Single-Token Dilemma:** Single-token systems fail to reconcile competing economic incentives: hard-capped assets are hoarded, crippling transactional velocity [1], while uncapped assets degrade purchasing power [2]. Conversely, stablecoins attempt to use external pegs but inevitably introduce custodial risk, suffer from severe capital inefficiency, or collapse under fragile algorithmic loops [3].
2. **Supply Entropy and the Precision Problem:** Unrecoverable coin loss permanently shrinks strictly capped supplies [4, 5]. Over time, the atomic base unit becomes too valuable, causing standard percentage fees to calculate into unprocessable fractions. Expanding decimal precision to fix this requires code migrations and raises costs because it increases storage and compute requirements.
3. **The Scalability vs Security Trade-Off:** Increasing transaction throughput on legacy networks typically compromises security or decentralisation [6]. Scaling horizontally (e.g., sharding or independent sidechains) fragments the network’s mining power, drastically lowering the cost to attack individual partitions [6]. Conversely, scaling vertically (expanding base-layer block capacity) centralises validation by restricting node operation to enterprise data centres, effectively reducing decentralisation [6].
4. **The Post-Quantum Horizon:** Legacy networks rely on Elliptic Curve Cryptography, which is mathematically vulnerable to Shor’s algorithm on scaled quantum hardware [7]. To survive, these established networks will eventually be forced to attempt retroactive state migrations to post-quantum standards like NIST FIPS 204 [8]. However, upgrading a live, multi-billion-dollar ledger after the fact is a monumental unresolved conflict: it requires highly contentious hard forks, risks splitting network consensus, and threatens to permanently strand vulnerable funds in legacy wallets that fail to migrate.

To address these tensions, this paper proposes Silicoin: an architecture that separates store of value from transactional utility at Layer 0, and models continuous physical commodity extraction via a permanent 0.2 Gold tail emission. It secures state transitions natively using post-quantum cryptographic standards. Rather than artificially restricting base-layer access, Silicoin relies on natural fee-market incentives to cleanly offload high-frequency retail and commercial execution to sovereign child chains (LDKs), preserving Layer 0 purely for decentralised settlement.

2 Layer 0: The Dual-Token Economy

2.1 The Nakamoto Foundation

Conceptually, Silicoin relies on the proven mechanics of Proof-of-Work (PoW) consensus to establish an immutable and decentralised root ledger [9]. Rather than restricting this base layer to a single asset, Silicoin natively issues two distinct assets side by side during block production:

Gold and Cash. The protocol enforces no exchange rate between them; it leaves convertibility entirely to the free, peer-to-peer market. Both assets are denominated in 10^8 atomic units (satoshis). By cleanly decoupling wealth preservation from transactional utility at the Layer 0 root, Silicoin structurally addresses the single-token dilemma.

2.2 Silicoin Gold: Store of Value

Silicoin Gold provides the inelastic scarcity required for long-term wealth preservation and pristine collateral. It mimics the emission mechanics of legacy hard-capped assets, following a standard schedule that begins with a block subsidy of 50 Gold and halves every 210,000 blocks [9]. While some systems assume a zero-block-reward environment will eventually fail, a fee-only market can theoretically sustain network security. As profitability drops, excess miners exit to reduce competition, or large stakeholders simply operate nodes at a minor loss to secure their own stored wealth. However, a strictly capped supply still suffers from permanent supply entropy, as coins are constantly lost due to various reasons.

To accommodate this continuous decay, Silicoin enforces a permanent tail floor of 0.2 Gold per block. Because coin loss is proportional to total supply, a fixed block reward converges to a stable monetary supply that is neither inflationary nor deflationary. In the long run, this microscopic tail emission acts as a thermodynamic offset, creating an equilibrium where new coins are created just as fast as they are lost [10].

2.3 Silicoin Cash: Medium of Exchange

Rather than relying on external fiat pegs or vulnerable algorithmic price controllers, Silicoin Cash is designed to expand and contract organically based on market demand. This floating medium of exchange is governed by three base-layer mechanisms:

- **Baseline Mint:** Every Layer 0 block mints a static baseline of 50 Cash to guarantee continuous network liquidity.
- **Permanent Activity Burns:** To organically counter supply expansion as network activity increases, a fixed percentage (5%) of all Layer 0 transaction fees is permanently burned. While variable-fee burn mechanisms have been proposed to manage blockchain congestion [11], their algorithmic feedback loops remain vulnerable to miner manipulation and simulated congestion. Silicoin’s static percentage relies on strict mathematics, penalising artificial block-stuffing and eliminating gameable exploits. Additionally, half (50%) of all child-chain (LDK) keep-alive fees are destroyed.
- **Dutch Auction Bond Sinks (Native Yield):** When market participants seek to absorb excess Cash liquidity or acquire future protocol yield, they participate in an automated, oracle-free bond auction running in 2,016-block epochs [12].

To ensure an active secondary market and prevent stalled liquidity traps, the bond market operates via discrete epoch benchmarks and a responsive retargeting formula:

- **Hard Quota & Block-Level Drip:** Each epoch allows a maximum issuance of 0.05% of the circulating Cash supply ($B_{\max}$). To prevent capital-rich actors from monopolising an epoch’s

supply in a single transaction (epoch sniping), this cap is dripped linearly across the epoch as a strict block-level quota ($b_{\text{block}} = \lceil B_{\text{max}}/2016 \rceil$). Any unabsorbed quota permanently expires at the end of the epoch, ensuring that low demand results in proportionally low or zero issuance.

- **Asymmetric Retargeting:** Each epoch has a fixed base ask price $P_e \in [70, 98]$. At the end of the epoch, the protocol evaluates the Utilisation Rate ($U_e = \frac{B_{\text{sold}}}{B_{\text{max}}}$) against a target equilibrium of 50% ($U^* = 0.5$). To prevent oscillating “ping-pong” attacks by large capital coordinators, the adjustment (ΔP) is asymmetric: high demand raises the price by up to +2 Cash, but low demand lowers the price by a maximum of only −1 Cash. The reserve price for the next epoch, P_{e+1} , is calculated as:

$$\Delta P = \begin{cases} +2 \times (2U_e - 1) & \text{if } U_e \geq 0.5 \\ +1 \times (2U_e - 1) & \text{if } U_e < 0.5 \end{cases}$$

$$P_{e+1} = \text{clamp}(P_e + \Delta P, 70, 98)$$

This ensures that while the protocol swiftly cuts yields during liquidity gluts, it requires prolonged, organic low demand to significantly raise yields. The genesis starting price is set near par at $P_0 = 95$ Cash.

- **Priority Bidding:** When demand is high and a block’s specific quota is contested, bids are prioritised by surplus burn (burning more than P_e), then transaction fee, and finally transaction ID.
- **Secondary Market Transferability:** Bond receipts are issued as transferable Layer 0 UTXOs. This enables a continuous secondary market where traders can reprice bonds in real time. If macroeconomic conditions shift between the 2,016-block protocol adjustments, secondary trading creates continuous price discovery and allows effective interest rates to trail inflation organically.
- **Maturity & Non-Additive Payouts:** Bonds mature after 20,160 blocks (approximately 10 weeks). Upon maturity, the 100 Cash face value is distributed as 98 Cash to the bondholder and 2 Cash to the miner as a dividend. Crucially, these payouts are non-additive; maturing bonds directly absorb the baseline 50-Cash block reward.

3 Scaling via Layer Development Kits (LDKs)

Routing every commercial transaction, smart contract, and application through a global base ledger creates severe throughput bottlenecks. Standard attempts to increase speed often sacrifice decentralisation by requiring enterprise-grade hardware or fragmenting mining power across vulnerable sidechains and shards. Silico addresses this by decoupling state execution from base-layer settlement via Layer Development Kits (LDKs). This architecture allows developers to launch sovereign child blockchains anchored directly to Layer 0, scaling throughput dramatically without fragmenting the network’s underlying Proof-of-Work security.

3.1 State Compartmentalisation

LDKs function as isolated execution environments for the high-velocity economy. Developers can deploy custom tokens, decentralised exchanges, or high-throughput applications within a sovereign child chain because application state is strictly compartmentalised; failed projects or high-volume contracts never bloat the main Layer 0 ledger. Rather than artificially restricting base-layer access, Silicoin relies on natural fee-market incentives to seamlessly route retail and high-frequency commercial execution into these efficient lanes, preserving Layer 0 purely for decentralised settlement.

3.2 Pristine Collateral and Local Execution

To establish a sovereign child chain and prevent double-spending, operators and users lock Gold or Cash into a dedicated Layer 0 escrow. In this context, Layer 0 acts as foundational commercial real estate: locking pristine collateral is the prerequisite for hosting a secure, high-volume settlement lane. Once capital is bridged, users transact at local, sub-second speeds. Crucially, child chains strictly execute state transitions and cannot mint native Gold or Cash. This renders Sybil attacks on child networks economically fruitless. If an LDK is attacked or its operator goes offline, users can unilaterally exit and unlock their funds on Layer 0 by providing a Merkle proof of their state.

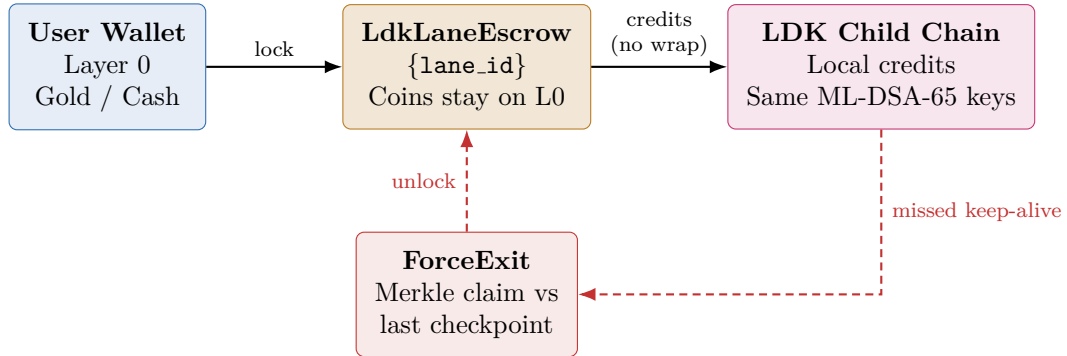


Figure 1: Layer 0 real-estate escrow flow. Native Gold or Cash is locked into an **LdkLaneEscrow** UTXO; the coins never leave Layer 0. The child lane issues local credits against that escrow under the same ML-DSA-65 keys (no wrapped tokens). If the operator misses a keep-alive, **ForceExit** recovers native funds from escrow via a Merkle proof against the last checkpoint.

3.3 Cryptographic Checkpoints

Child chains regularly settle back to Layer 0, relying on the base ledger as an absolute, decentralised clock. Each checkpoint contains:

- A succinct cryptographic proof (utilising Winterfell STARKs) validating local state arithmetic.
- Full state diffs published directly to Layer 0 to ensure data availability. This guarantees that any Layer 0 node can independently reconstruct and verify the child state without trusting the LDK operator.

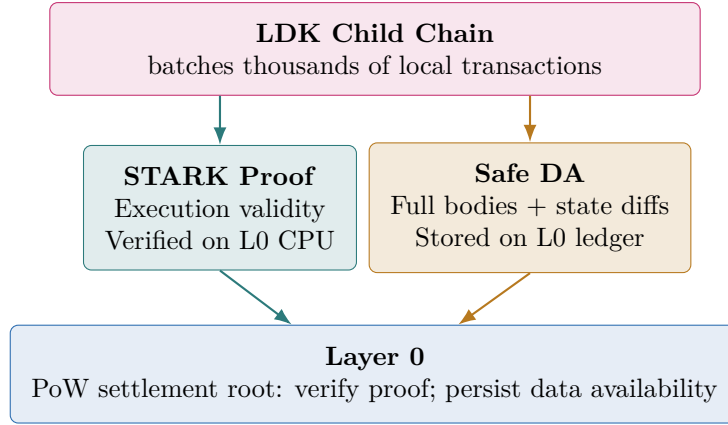


Figure 2: Checkpoints and STARK data flow. Each settlement carries both a Winterfell STARK (execution validity, verified natively on Layer 0 CPUs) and Safe DA payloads (full settlement bodies and typed state diffs stored on the Layer 0 ledger so users can reconstruct state and force an exit).

3.4 LDK Lifecycle and Keep-Alives

To ensure abandoned child networks do not perpetually congest Layer 0, Silicoins enforces strict lifecycle management. Active LDKs must submit keep-alive transactions every 2,016 blocks. If an LDK misses this window, the protocol automatically deactivates its slot. Keep-alive fees follow a dedicated thermodynamic split: 50% is permanently burned (acting as a deflationary sink for the Cash supply), and 50% is awarded to the Layer 0 miner to incentivise continuous base-layer security.

3.5 Nested Hierarchy and Fee Autonomy

Silicoins's architecture supports fractal, multi-layered nesting. Child lanes can nest further, creating arbitrarily deep Layer N networks that anchor upward through an intermediate parent before ultimately settling on Layer 0.

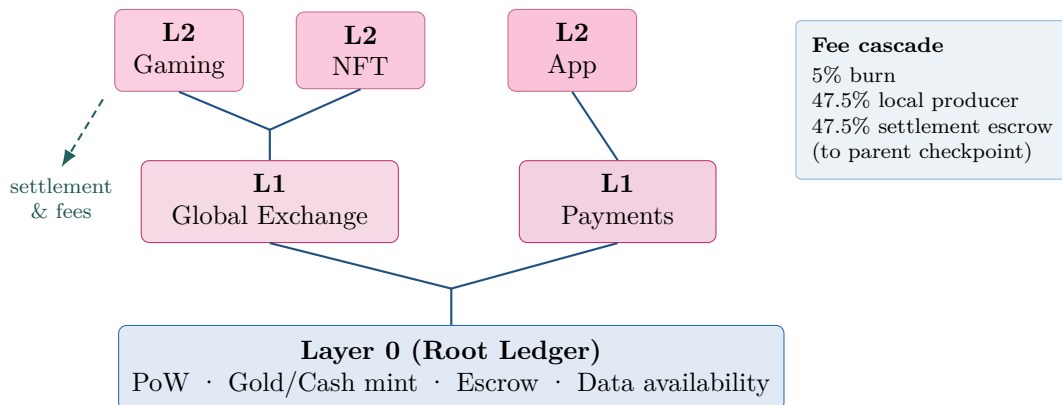


Figure 3: The fractal LDK tree and fee cascade. Layer 0 sits at the absolute bottom as the root ledger. Branches route upward to L1 LDKs (e.g. a global exchange) and further into L2 ecosystems. Downward settlement follows the reference fee cascade: 5% burned locally, 47.5% to the local block producer, and 47.5% into settlement escrow to fund the parent checkpoint.

The Fee Cascade: As checkpoints route upward, they follow a designated fee cascade. For the reference L1 LDK, child-lane block fees follow a 5 / 47.5 / 47.5 split (5% burned, 47.5% to the settlement escrow, and 47.5% to the local block producer).

LDK Sovereignty: Layer 0 cryptographically verifies checkpoints and escrow funding but explicitly does not dictate child-lane monetary policy; independent operators are entirely free to establish custom internal fee markets.

Anti-Stall Mechanisms: To prevent nested networks from stalling during Layer 0 congestion, child sequencers can utilise Cash-denominated Replace-By-Fee (RBF) and CPFP-lite mechanisms to outbid base-layer mempool spam.

3.6 Scripting Expandability

In addition to STARK-based LDKs, Silicoin features a native, highly secure scripting language designed to accommodate alternative off-chain scaling topologies, including the potential future deployment of Lightning-adjacent state channel networks.

4 Security: Post-Quantum Cryptography

As quantum computing hardware advances, legacy cryptographic algorithms, specifically elliptic curves like ECDSA used in legacy networks, face an imminent threat of decryption via Shor’s algorithm. Most existing blockchain networks treat post-quantum migration as a future hard-fork problem, leaving their genesis state permanently exposed. Silicoin integrates post-quantum cryptography natively at the protocol level from day one:

- **ML-DSA-65 Signatures:** All consensus-level transaction authorisations across Layer 0 utilise ML-DSA-65 (Module-Lattice-Based Digital Signature Standard). This FIPS 204 lattice-based scheme resists both classical and quantum computing attacks.
- **STARK Proof Integrity:** Child networks utilise Winterfell STARK (Scalable Transparent Arguments of Knowledge) proofs to validate local state execution. STARKs provide transparent, post-quantum secure cryptographic receipts without requiring trusted setups.
- **Layer 0 Hybrid Execution:** Raw ML-DSA signatures are verified natively on Layer 0 CPUs. This deliberately bypasses the extreme computational overhead that would occur if complex lattice cryptography had to be verified directly inside zero-knowledge circuits.

5 Limitations & Future Work

While Silicoin introduces robust theoretical mechanisms to address the single-token dilemma and scalability bottlenecks, several parameters and assumptions require real-world validation:

- **Unit Bias:** A common psychological barrier in cryptocurrency is “unit bias,” where users prefer transacting in whole numbers rather than fractions. As prior research shows, standardising the “satoshi” (10^{-8}) as the default wallet display unit does not effectively solve

this issue. Silicoins currently does not natively alter its base mechanics to address unit bias, leaving it as an unresolved protocol dynamic.

- **Efficacy of the Permanent Tail:** The 0.2 Gold emission floor is designed to offset permanent coin loss, a mechanism adapted from single-token systems. However, its utility in a dual-token architecture is questionable. In a system like Bitcoin, unrecoverable coin loss eventually cripples its utility as a currency because the atomic base unit becomes too valuable for ordinary, small-scale transactions. Silicoins Gold, however, is strictly a store of value used for large-scale wealth preservation. Because Gold transactions are inherently large, extreme fractionalisation is not a practical constraint. For a dedicated reserve asset, natural supply deflation via coin loss actually reinforces its scarcity and benefits holders. Therefore, using a tail emission to solve a medium-of-exchange problem on a pure store-of-value asset may prove entirely redundant.
- **Parameter Optimisation:** Throughout this paper, specific constants (e.g., the 5% activity burn) are used to illustrate the economic engine. Because hardcoded arbitrary values can create systemic fragility, these parameters must be rigorously modelled and optimised before mainnet launch.
- **Impact of the Non-Additive Baseline:** The protocol strictly requires maturing bonds to absorb the 50 Cash block baseline to prevent arbitrary supply inflation. A potential limitation is whether a 50 Cash baseline is large enough to matter at meaningfully high market capitalisations. However, enforcing this invariant remains mathematically necessary to guarantee a strict supply ceiling during periods of high bond redemption.

6 Conclusion

Current single-token architectures struggle to reconcile competing economic incentives. Addressing this requires decoupling wealth preservation from daily commerce via a native dual-token economy. In doing so, Silicoins structurally resolves the single-token dilemma. Relying on centralised fiat collateral or external price oracles is no longer necessary. Organic market demand dictates the expansion and contraction of the Cash supply instead. Scaling is then handled horizontally via Layer Development Kits (LDKs). Without fragmenting the root network's Proof-of-Work security, high-throughput applications are isolated. Every child network settles back to Layer 0 using transparent cryptographic proofs. Ensuring long-term viability, ML-DSA-65 signatures secure these base-layer state transitions from day one. Together, these mechanisms provide a robust, future-proof blueprint for a self-sustaining decentralised economy.

References

- [1] D. Yermack, “Is Bitcoin a Real Currency? An Economic Appraisal,” in *Handbook of Digital Currency*, Elsevier, 2015, pp. 31–43.
- [2] W. Bolt and M. R. van Oordt, “On the Value of Virtual Currencies,” *Journal of Money, Credit and Banking*, vol. 52, no. 4, pp. 835–862, 2020.
- [3] C. Catalini and A. de Gortari, “On the Economic Design of Stablecoins,” Social Science Research Network (SSRN), Aug. 2021.
- [4] J. J. Roberts and N. Rapp, “Exclusive: Nearly 4 Million Bitcoins Lost Forever, New Study Says,” *Fortune*, Nov. 25, 2017. [Online]. Available: <https://fortune.com/2017/11/25/lost-bitcoins/>
- [5] Coin Metrics, “How Many Bitcoins Are Permanently Lost?,” *State of the Network: Issue 26*, Nov. 19, 2019. [Online]. Available: <https://coinmetrics.substack.com/p/coin-metrics-state-of-the-network-d2e>
- [6] S. Mssassi and A. Abou El Kalam, “The Blockchain Trilemma: A Formal Proof of the Inherent Trade-Offs Among Decentralisation, Security, and Scalability,” *Applied Sciences*, vol. 15, no. 1, p. 19, Dec. 2024.
- [7] P. W. Shor, “Algorithms for quantum computation: discrete logarithms and factoring,” *Proceedings 35th Annual Symposium on Foundations of Computer Science*, IEEE, 1994, pp. 124–134.
- [8] National Institute of Standards and Technology (NIST), “FIPS 204: Module-Lattice-Based Digital Signature Standard,” U.S. Department of Commerce, 2024.
- [9] S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” <http://www.bitcoin.org>, 2008.
- [10] P. Todd, “Surprisingly, Tail Emission Is Not Inflationary,” Peter Todd’s Blog, Jul. 9, 2022.
- [11] T. Roughgarden, “Transaction Fee Mechanism Design for the Ethereum Blockchain: An Economic Analysis of EIP-1559,” ACM Conference on Economics and Computation, 2020.
- [12] C. C. Moallemi and D. Robinson, “Loss-Versus-Fair: Efficiency of Dutch Auctions on Blockchains,” arXiv preprint arXiv:2406.00113v2, 2024.